

Limits of Computation



Hunting a Frog

Frog picks a number $k > 0$. That becomes the frog's strategy!

At time 1, the frog jumps to k

At time 2, the frog jumps to $2k$

...



$$k = 3$$

Hunting a Frog

Frog picks a number $k > 0$. At time t , the frog jumps to $tk!$



$$k = 3$$

Hunting a Frog

Frog picks a number $k > 0$. At time t , the frog jumps to $tk!$



Now comes a hunter!



$$k = 3$$

Hunting a Frog

Frog picks a number $k > 0$. At time t , the frog jumps to $tk!$

"There is a pre-determined strategy with which I can shoot, so that the frog dies."



Strategy of Squares

Frog picks a number $k > 0$. At time t , the frog jumps to $tk!$

At time t , the hunter shoots at t^2

If the frog chose k , it will be
shot at time k at position k^2



Strategy of Squares

Even after knowing the strategy of squares,
why can the frog not outsmart the hunter?



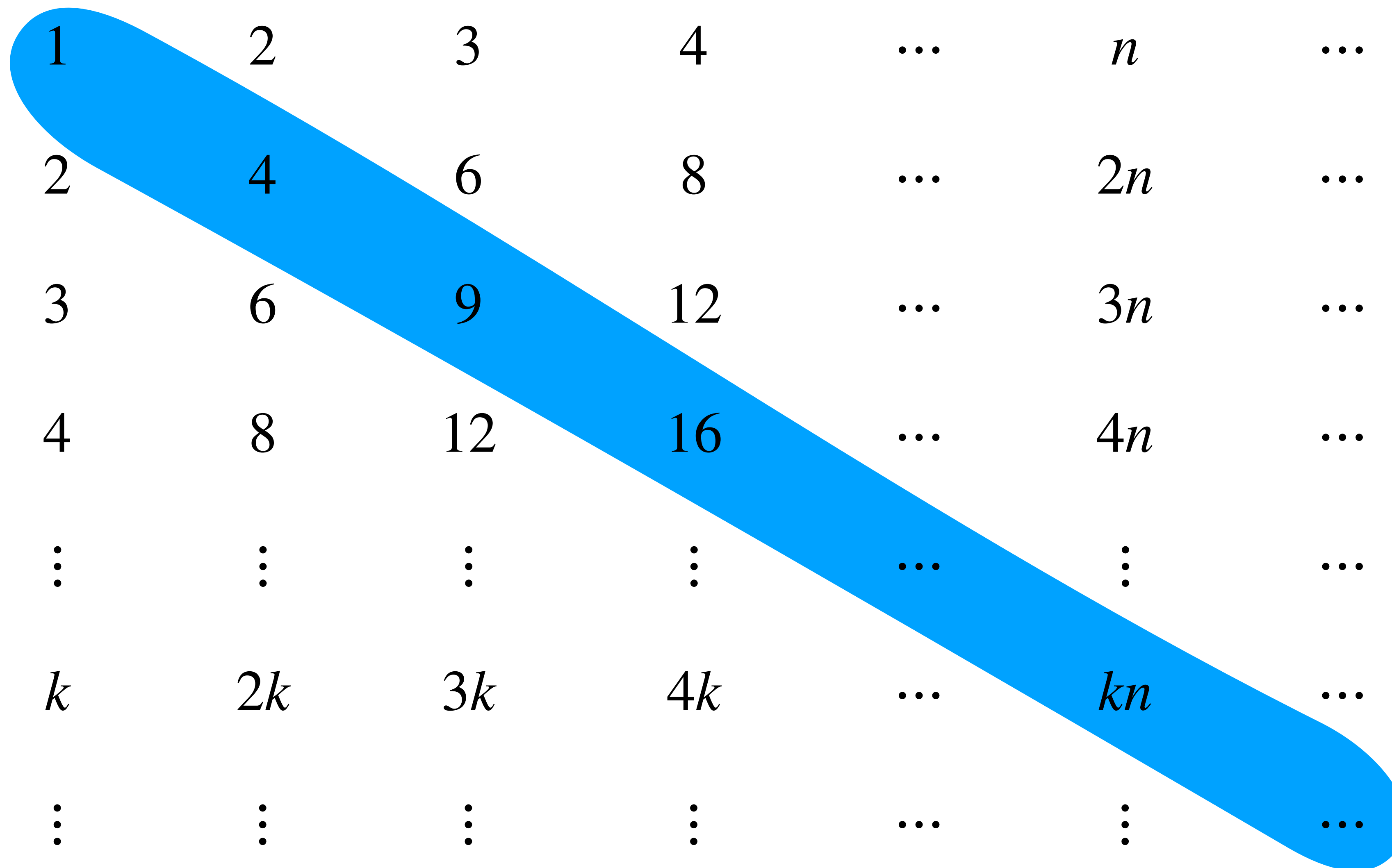
Frog Strategy

Time

1	2	3	4	...	n	...
2	4	6	8	...	$2n$	...
3	6	9	12	...	$3n$	...
4	8	12	16	...	$4n$	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	...	$\vdots$	...
k	$2k$	$3k$	$4k$	...	kn	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	...	$\vdots$	...

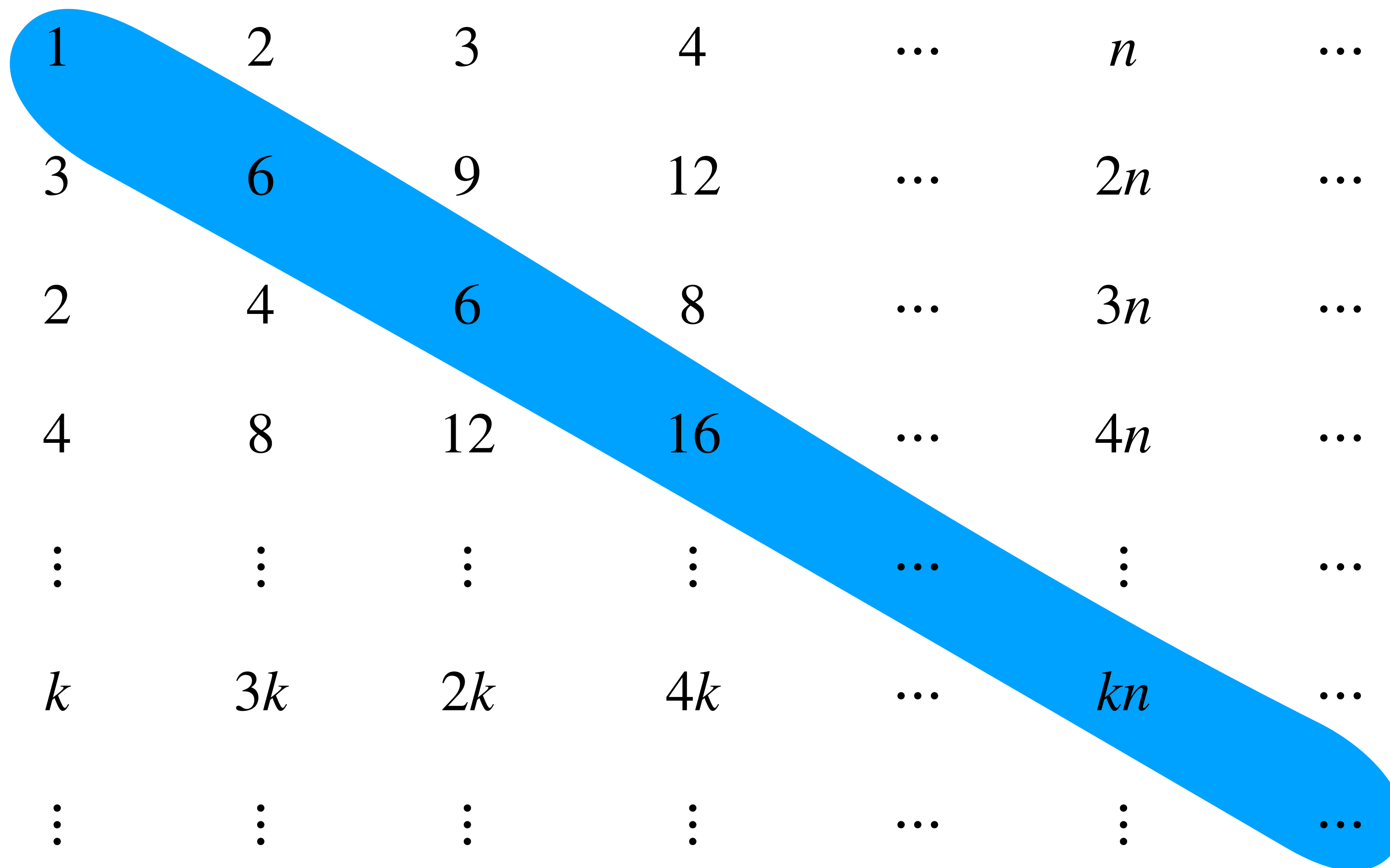
Frog Strategy

Time



Frog Strategy

Time



Frogs with Fractions!

Frog picks a number $k > 0$. At time t , the frog jumps to tk !

This number can be any fraction.

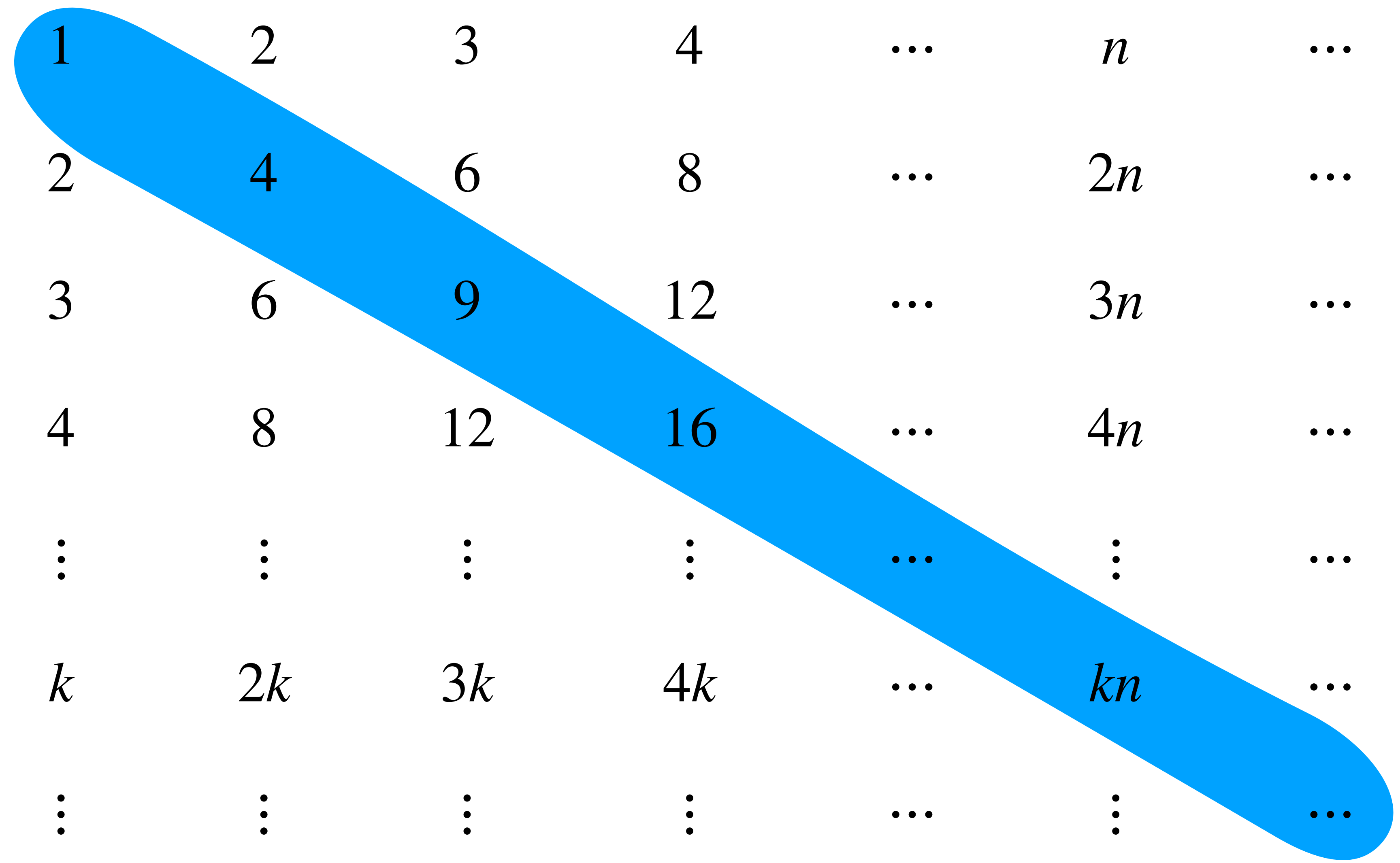
Does the hunter have a strategy?



Frog Strategy

$p \mid q$

Time



Can I enumerate fractions of the form p/q ?

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$\frac{1}{1}$	$\frac{2}{1}$	$\frac{3}{1}$	$\frac{4}{1}$	$\dots$	$\frac{n}{1}$	$\dots$
$\frac{1}{2}$	$\frac{2}{2}$	$\frac{3}{2}$	$\frac{4}{2}$	$\dots$	$\frac{n}{2}$	$\dots$
$\frac{1}{3}$	$\frac{2}{3}$	$\frac{3}{3}$	$\frac{4}{3}$	$\dots$	$\frac{n}{3}$	$\dots$
$\frac{1}{4}$	$\frac{2}{4}$	$\frac{3}{4}$	$\frac{4}{4}$	$\dots$	$\frac{n}{4}$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\dots$
$\frac{1}{k}$	$\frac{2}{k}$	$\frac{3}{k}$	$\frac{4}{k}$	$\dots$	$\frac{n}{k}$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\dots$

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$\frac{1}{3}$	$\frac{2}{3}$	$\frac{3}{3}$	$\frac{4}{3}$	$\dots$	$\frac{n}{3}$	$\dots$
$\frac{1}{4}$	$\frac{2}{4}$	$\frac{3}{4}$	$\frac{4}{4}$	$\dots$	$\frac{n}{4}$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\dots$
$\frac{1}{k}$	$\frac{2}{k}$	$\frac{3}{k}$	$\frac{4}{k}$	$\dots$	$\frac{n}{k}$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\dots$

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$\frac{1}{2}$	$\frac{2}{2}$	$\frac{3}{2}$	$\frac{4}{2}$	$\dots$	$\frac{n}{2}$	$\dots$
$\frac{1}{3}$	$\frac{2}{3}$	$\frac{3}{3}$	$\frac{4}{3}$	$\dots$	$\frac{n}{3}$	$\dots$
$\frac{1}{4}$	$\frac{2}{4}$	$\frac{3}{4}$	$\frac{4}{4}$	$\dots$	$\frac{n}{4}$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\dots$
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$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\dots$

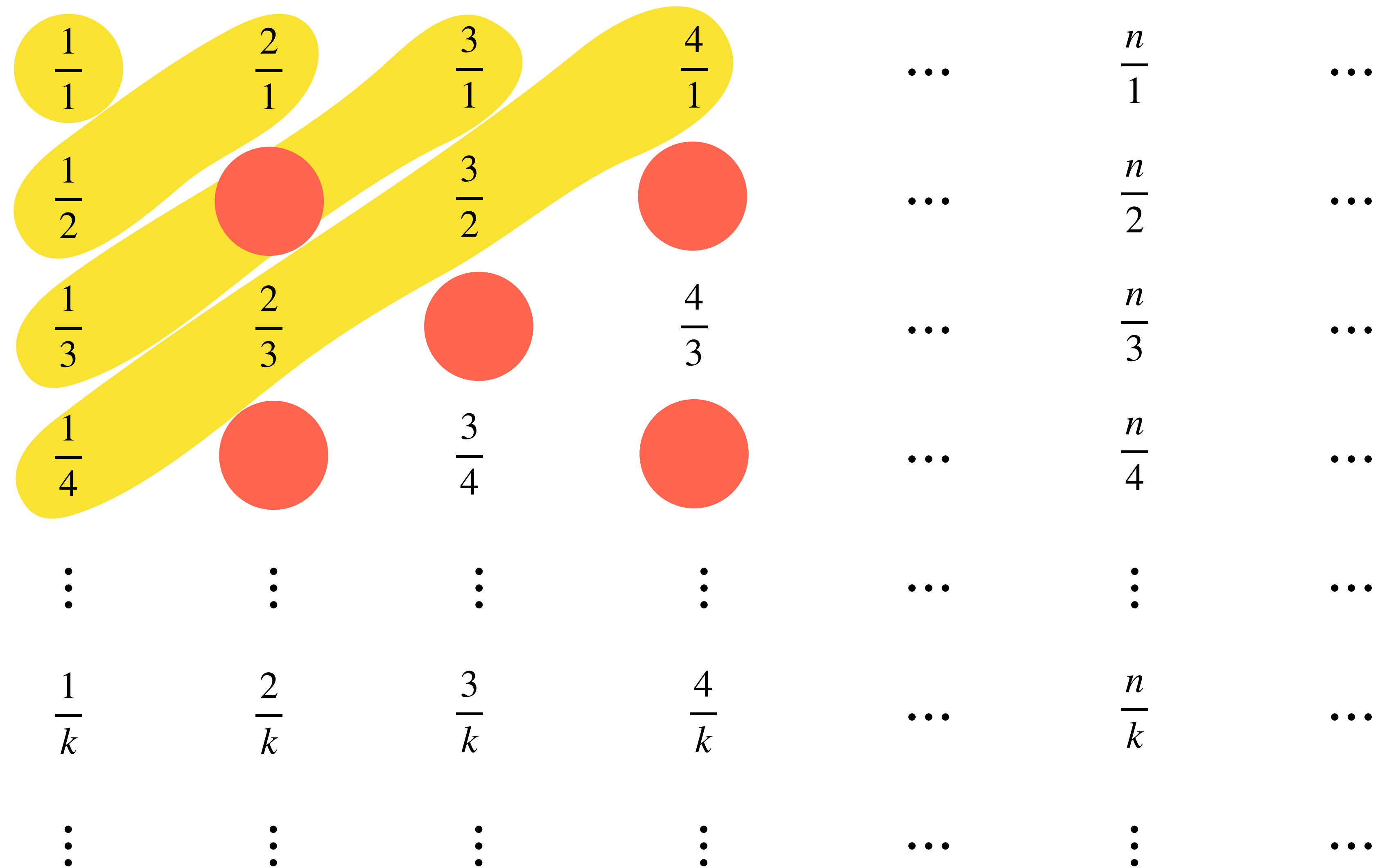
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$\frac{1}{3}$	$\frac{2}{3}$	$\frac{3}{3}$	$\frac{4}{3}$	$\dots$	$\frac{n}{3}$	$\dots$
$\frac{1}{4}$	$\frac{2}{4}$	$\frac{3}{4}$	$\frac{4}{4}$	$\dots$	$\frac{n}{4}$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\dots$
$\frac{1}{k}$	$\frac{2}{k}$	$\frac{3}{k}$	$\frac{4}{k}$	$\dots$	$\frac{n}{k}$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\dots$

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$\frac{1}{3}$	$\frac{2}{3}$	$\frac{3}{3}$	$\frac{4}{3}$	...	$\frac{n}{3}$	...
$\frac{1}{4}$	$\frac{2}{4}$	$\frac{3}{4}$	$\frac{4}{4}$	...	$\frac{n}{4}$	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	...	$\vdots$	...
$\frac{1}{k}$	$\frac{2}{k}$	$\frac{3}{k}$	$\frac{4}{k}$	...	$\frac{n}{k}$	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	...	$\vdots$	...

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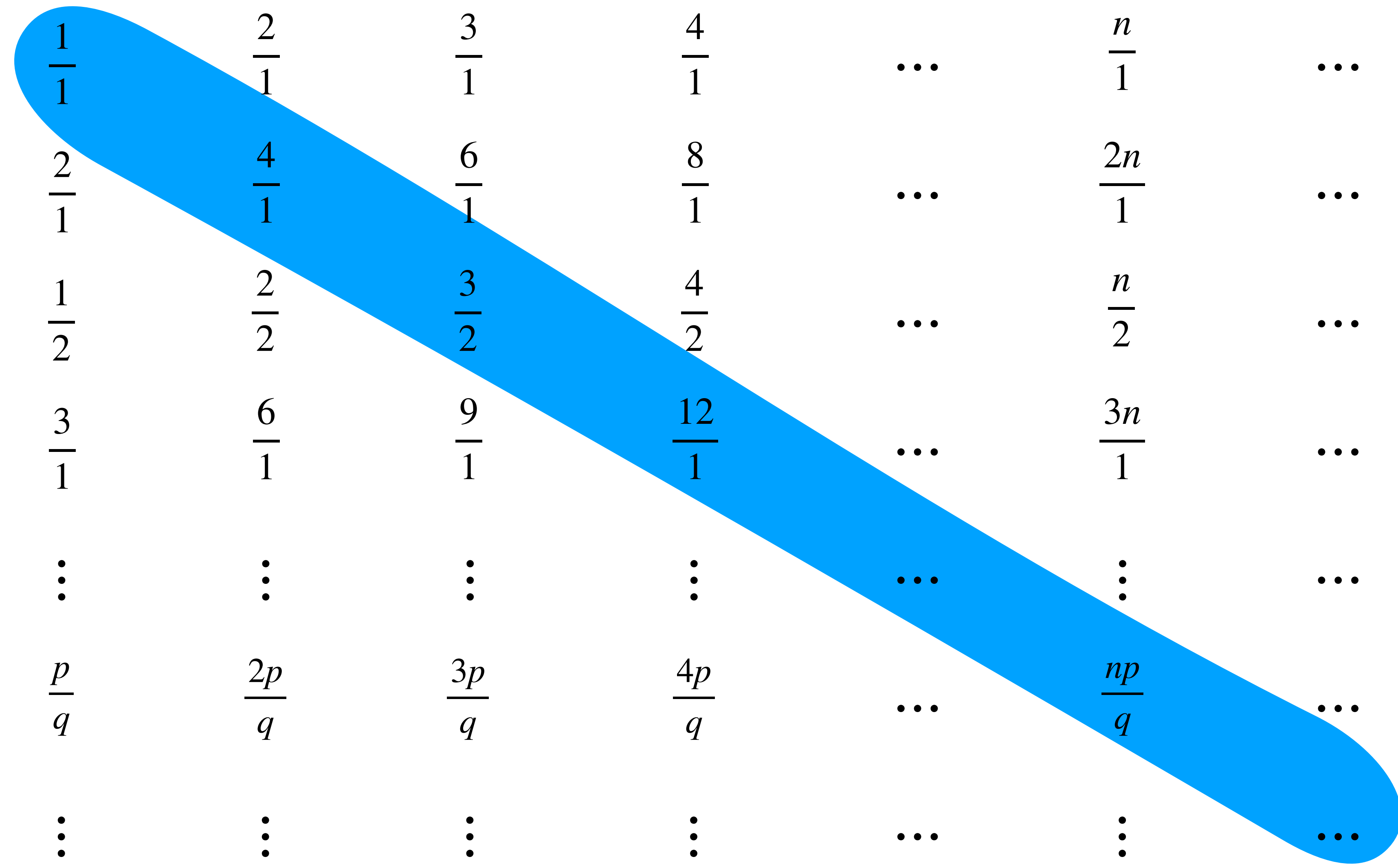


Can I enumerate fractions of the form p/q ?

$\frac{1}{1}$ $\frac{2}{1}$ $\frac{1}{2}$ $\frac{3}{1}$ $\frac{1}{3}$ $\frac{4}{1}$ $\frac{3}{2}$ $\frac{2}{3}$ $\frac{1}{4}$ $\dots$

Frog Strategy

Time



Big Tech Says Generative AI Will Save the Planet. It Doesn't Offer Much Proof

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Published: January 21, 2026 1.35pm GMT



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Harnessing AI to Help Save the Planet

INSIGHT OCTOBER 16, 2025



Can AI save the frog?



“My nose will grow!”



Grelling-Nelson Paradox

short
English
adjectival
pentasyllabic

autological

long
German
palindrome
monosyllabic

heterological

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heterological?

Berry Paradox

Each number can be described using a sentence in English.

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For example:

“The number of moons of the Earth”

“Number of students in this classroom”

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“Number whose digits are four one seven six two two three ...”

Berry Paradox

~~Each number can be described using a sentence in English.~~

Set of numbers that can be described in English using only 60 letters

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Consider the “smallest positive integer not definable in under sixty letters.”

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~~Each number can be described using a sentence in English.~~

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Consider the “smallest positive integer not definable in under sixty letters.”

Is it in the set?

Berry Paradox

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Set of numbers that can be described in English using only 60 letters

Consider the “smallest positive integer not definable in under sixty letters.”

“THE SMALLEST NUMBER THAT CANNOT BE DESCRIBED IN ENGLISH USING
LESS THAN SIXTY LETTERS”

Frogs with Real Numbers!

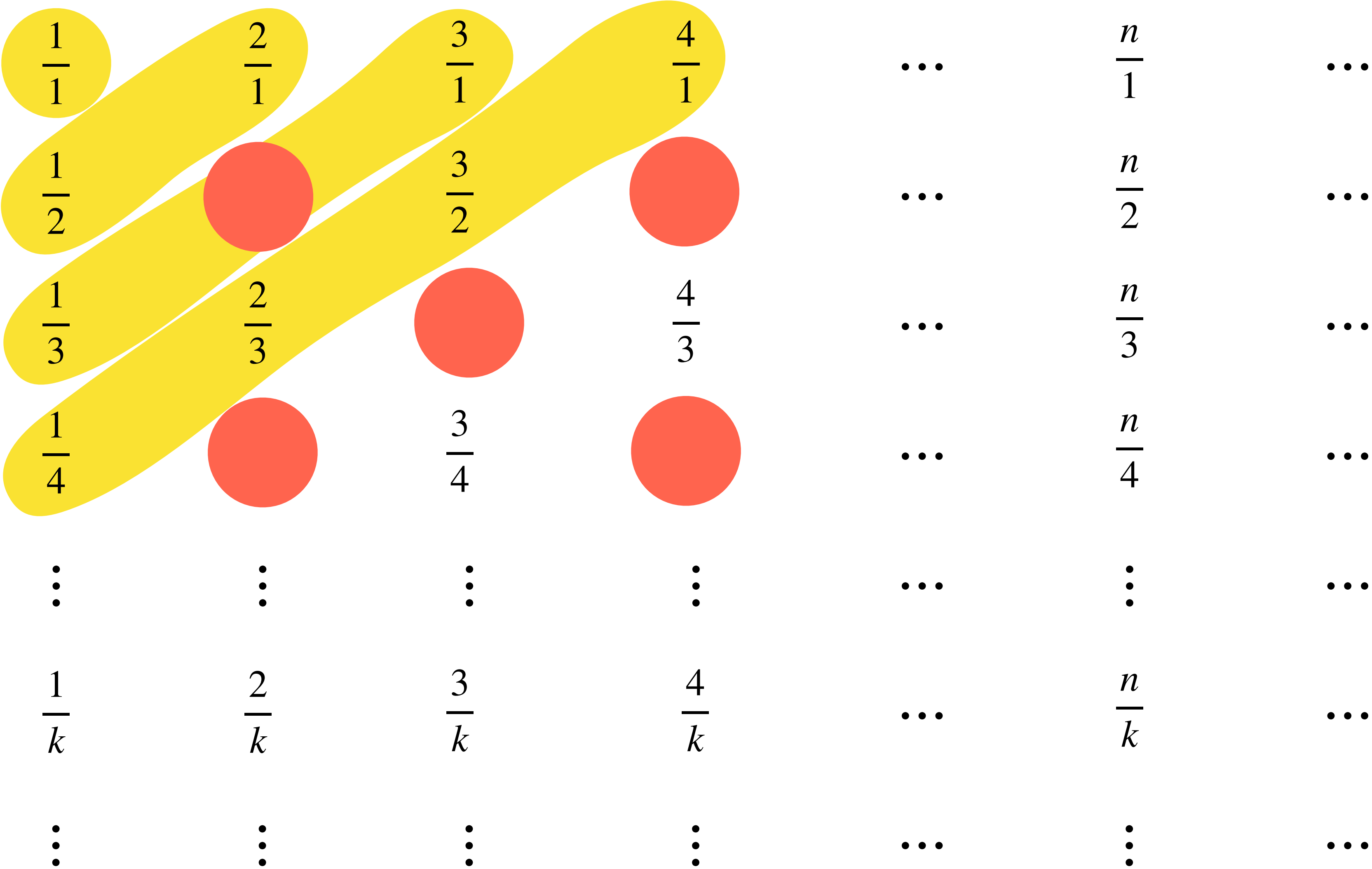
Frog picks a number $k > 0$. At time t , the frog jumps to tk !

This number can be any real number.

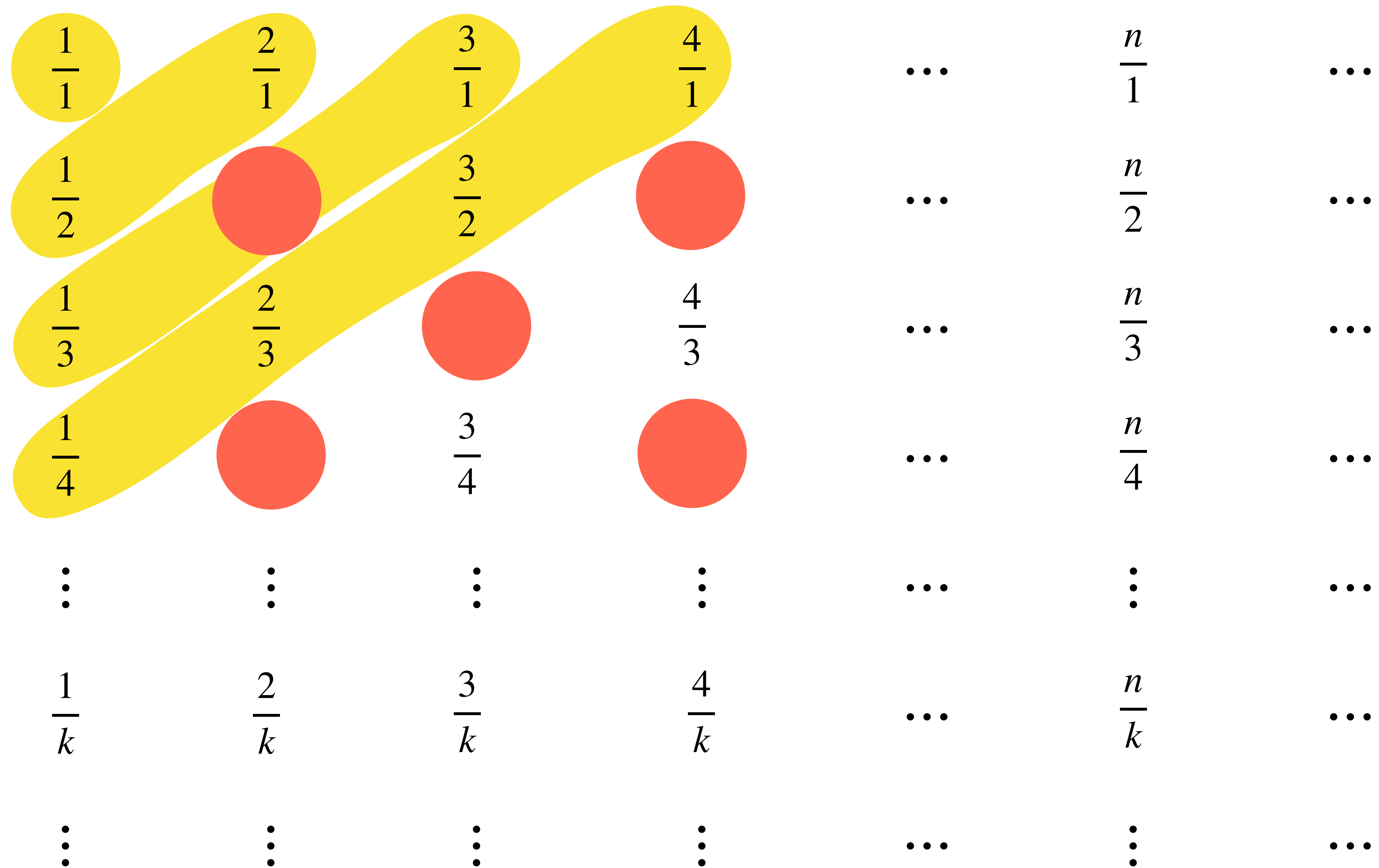
Does the hunter have a strategy?



Rational Numbers are Enumerable!



Are real numbers enumerable?



Are real numbers enumerable?

3.	1	4	1	...
1.	4	1	4	...
2.	0	0	0	...
2.	7	1	8	...
⋮	⋮	⋮	⋮	...

For sake of contradiction, suppose they were...

1	3.	1	4	1	...
2	1.	4	1	4	...
3	2.	0	0	0	...
4	2.	7	1	8	...
⋮	⋮	⋮	⋮	⋮	...

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⋮	⋮	⋮	⋮	⋮	...

We will construct a new real number that is not in the list...

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⋮	⋮	⋮	⋮	⋮	...

0.

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For sake of contradiction, suppose they were...

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3	2.	0	0	0	...
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⋮	⋮	⋮	⋮	⋮	...
	0.	2			

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⋮	⋮	⋮	⋮	⋮	...
	0.	2	2		

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⋮	⋮	⋮	⋮	⋮	...
	0.	2	2	1	

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It differs from the i^{th} number at the i^{th} place after the decimal point.

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	0.	2	2	1	...

It differs from the i^{th} number at the i^{th} place after the decimal point.

There are more reals than natural numbers!

There are more reals than natural numbers!

The number of rationals and natural numbers are the same.

There are more reals than natural numbers!

The number of rationals and natural numbers are the same.

Between every two rational numbers there is an irrational number.

Between every two irrational numbers there is an rational number.

And yet there are more irrational numbers than rational numbers!

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INSIGHT OCTOBER 16, 2025



Can AI save the frog?



**What can computer
science *never* do?**

Decision Problem: A computational problem that can be posed as a yes-no question about a given input.

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Decision Problem on Natural Numbers:

A problem that takes a natural number n as an input and returns yes or no as an output.

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Every problem can be solved, if and only if there is a program for each problem.

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Decision Problem on Natural Numbers:

A problem that takes a natural number n as an input and returns yes or no as an output.

Every problem can be solved, if and only if there is a program for each problem.

How many programs?

How many problems?

How many programs?

```
def is_even(number):  
    if number % 2 == 0:  
        return True  
    else:  
        return False
```

```
def is_prime(n):  
    if n <= 1:  
        return False  
    for i in range(2, int(n ** 0.5) + 1):  
        if n % i == 0:  
            return False  
    return True
```

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Each program is a string.

Each string is a number.

We can enumerate programs in increasing length.

How many programs?

N

How many programs?

N

How many problems?

is_less_than_4:



is_even:



is_prime:



is:



isn't:



0 `is_less_than_4:` 0 1 2 3 4 5 6 7 8 9 10 11 12 ...

1 `is_even:` 0 1 2 3 4 5 6 7 8 9 10 11 12 ...

2 `is_prime:` 0 1 2 3 4 5 6 7 8 9 10 11 12 ...

3 `is:` 0 1 2 3 4 5 6 7 8 9 10 11 12 ...

4 `isn't:` 0 1 2 3 4 5 6 7 8 9 10 11 12 ...

0 is_less_than_4:

0	1	2	3	4	5	6	7	8	9	10	11	12
---	---	---	---	---	---	---	---	---	---	----	----	----

 ...

1 is_even:

0	1	2	3	4	5	6	7	8	9	10	11	12
---	---	---	---	---	---	---	---	---	---	----	----	----

 ...

2 is_prime:

0	1	2	3	4	5	6	7	8	9	10	11	12
---	---	---	---	---	---	---	---	---	---	----	----	----

 ...

3 is:

0	1	2	3	4	5	6	7	8	9	10	11	12
---	---	---	---	---	---	---	---	---	---	----	----	----

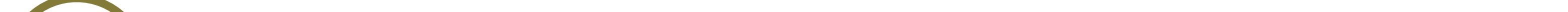
 ...

4 isn't:

0	1	2	3	4	5	6	7	8	9	10	11	12
---	---	---	---	---	---	---	---	---	---	----	----	----

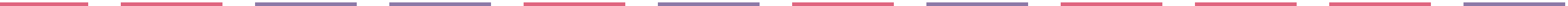
 ...

new:

0 is_less_than_4: 

1 `is_even:`

0	1	2	3	4	5	6	7	8	9	10	11	12	...
---	---	---	---	---	---	---	---	---	---	----	----	----	-----

2 is_prime:  ...

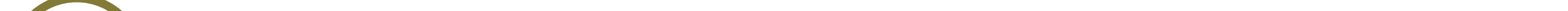
3 is:

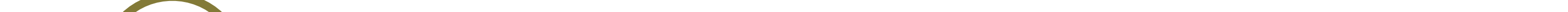
0	1	2	3	4	5	6	7	8	9	10	11	12	...
---	---	---	---	---	---	---	---	---	---	----	----	----	-----

4 isn't:

0	1	2	3	4	5	6	7	8	9	10	11	12	...
---	---	---	---	---	---	---	---	---	---	----	----	----	-----

new:

0 is_less_than_4: 

1 is_even:  ...

```
2  is_prime:  0  1  2  3  4  5  6  7  8  9  10 11 12  ...
```

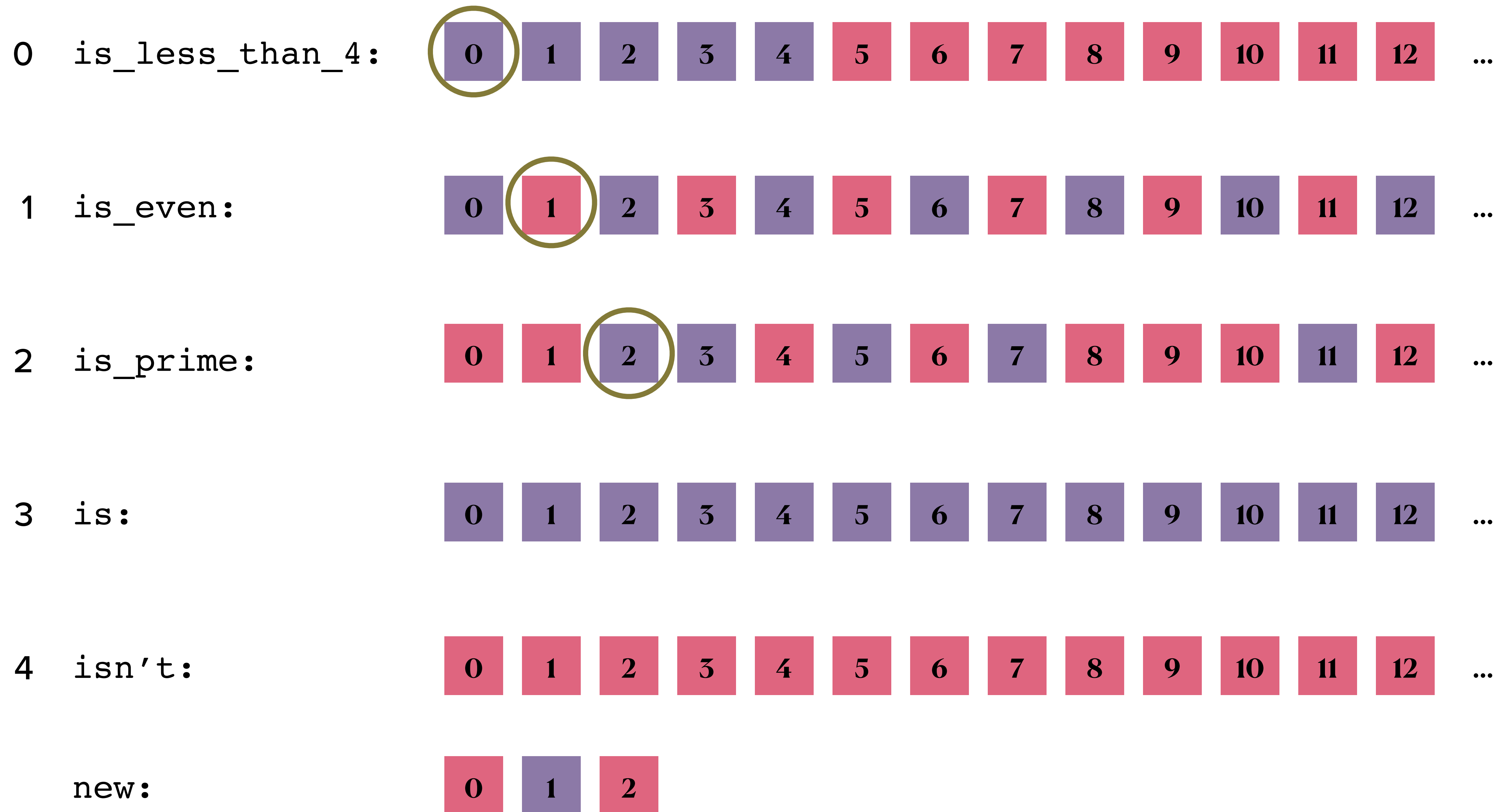
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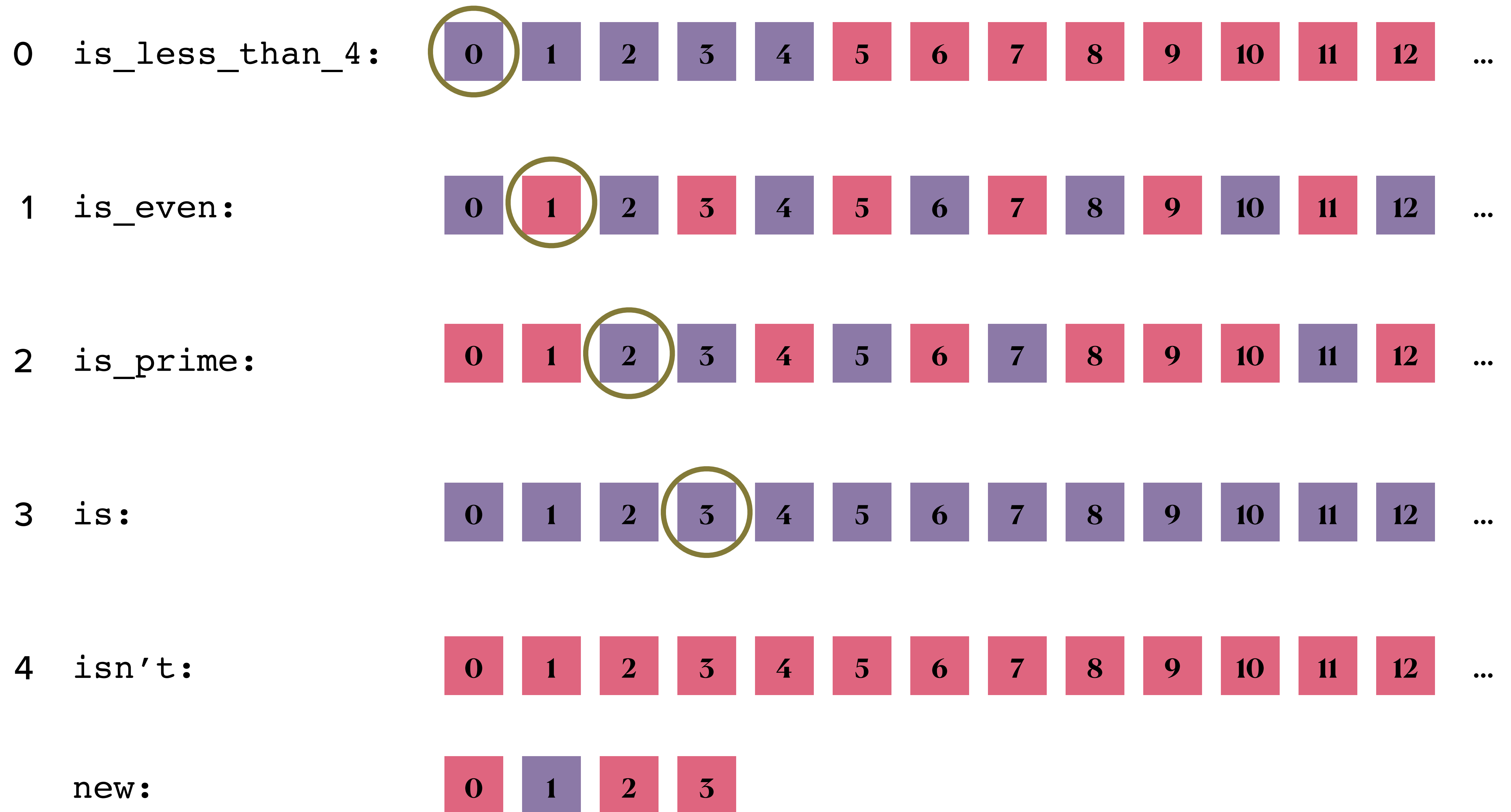
0	1	2	3	4	5	6	7	8	9	10	11	12	...
---	---	---	---	---	---	---	---	---	---	----	----	----	-----

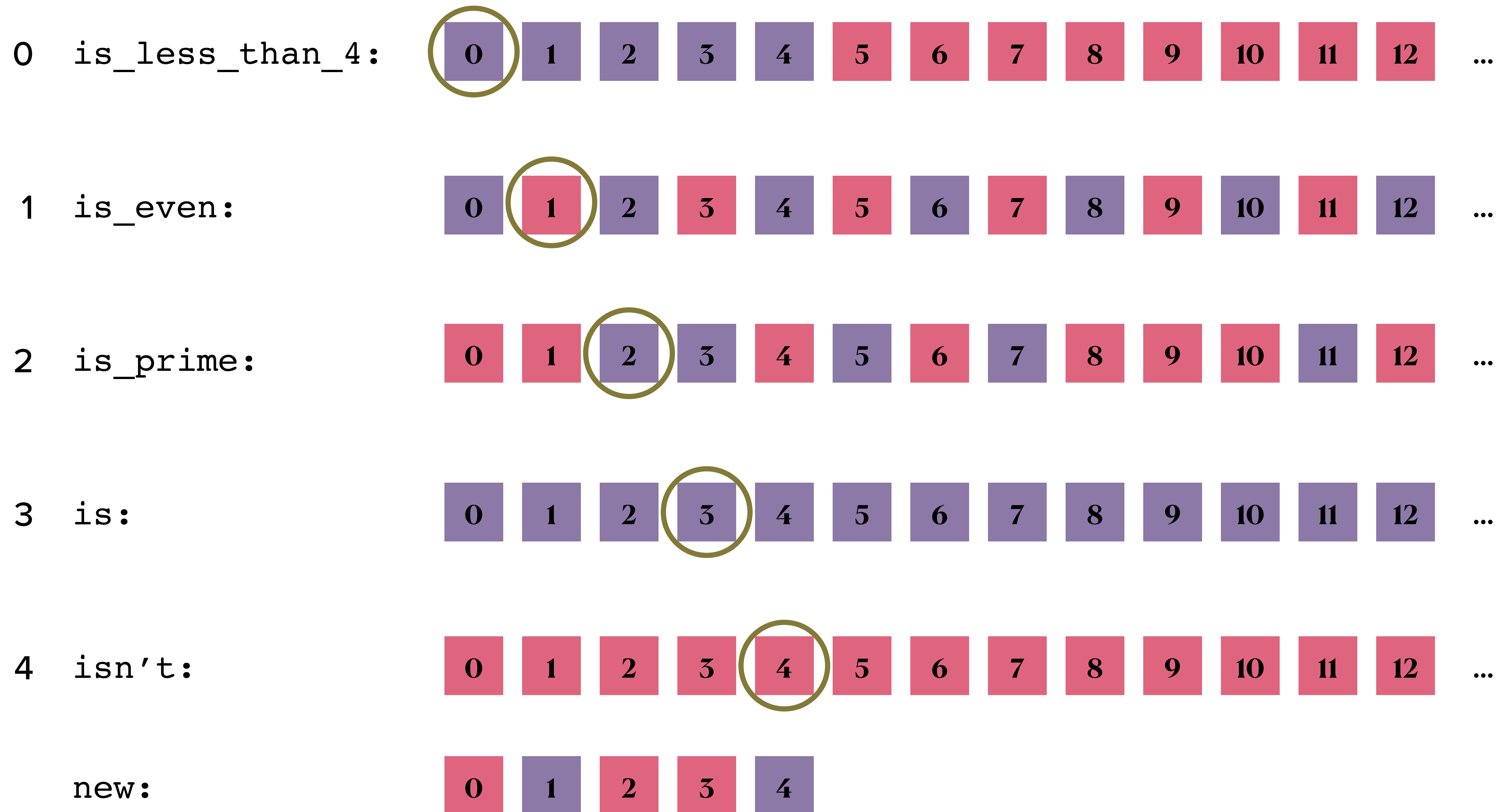
4 isn't:

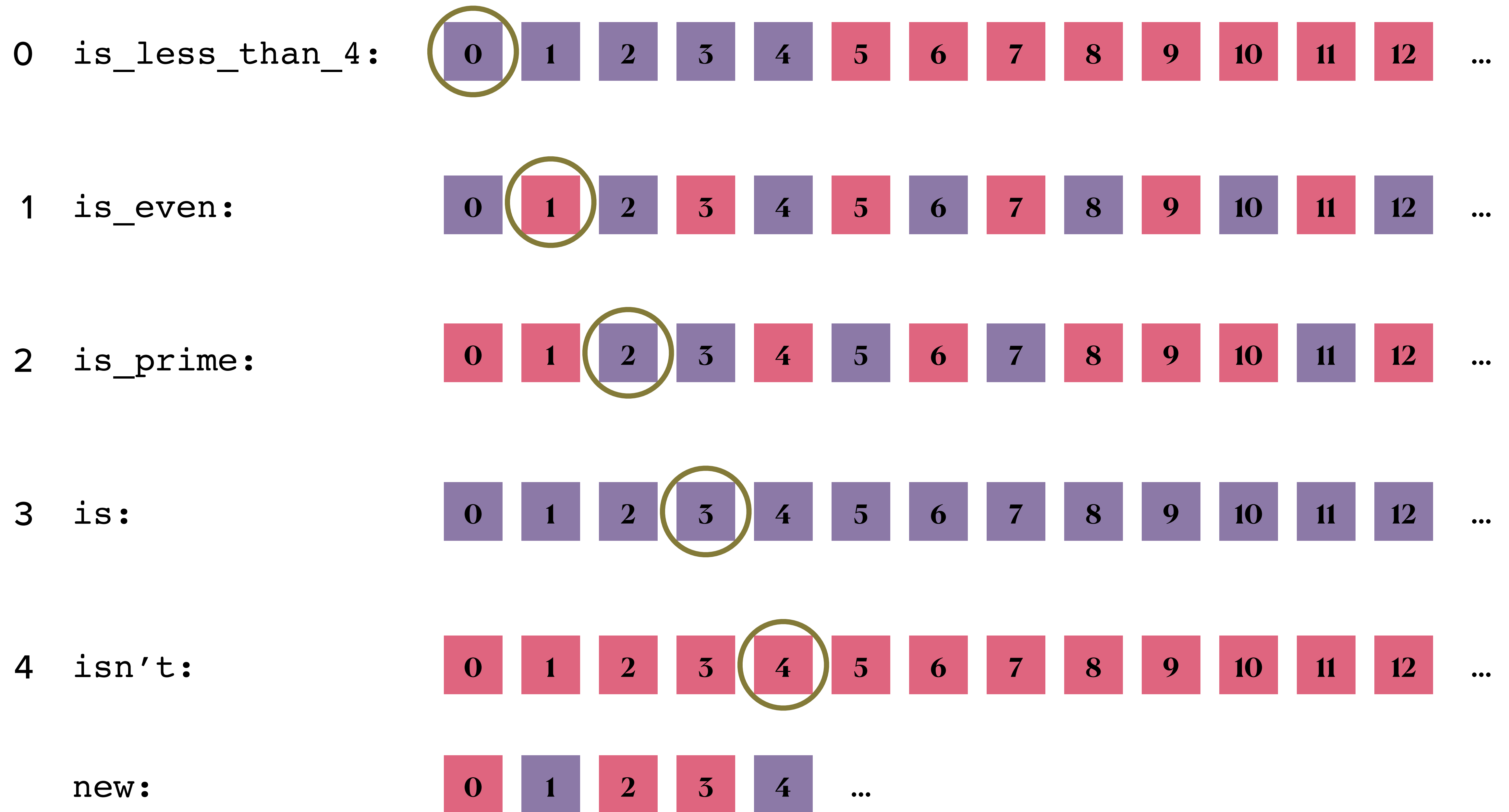
0	1	2	3	4	5	6	7	8	9	10	11	12	...
---	---	---	---	---	---	---	---	---	---	----	----	----	-----

new: 0 1









Why is new not in the enumeration of problems?

new:



Why is new not in the enumeration of problems?

new:



What if we had selected a different sequence for enumeration?

Computation and Its Limitations!

Frog picks any computable function $M : \mathbb{N} \rightarrow \mathbb{N}$.

At time t , frog jumps to $M(t)$

We have a fixed hunting strategy that works regardless of what the frog does.



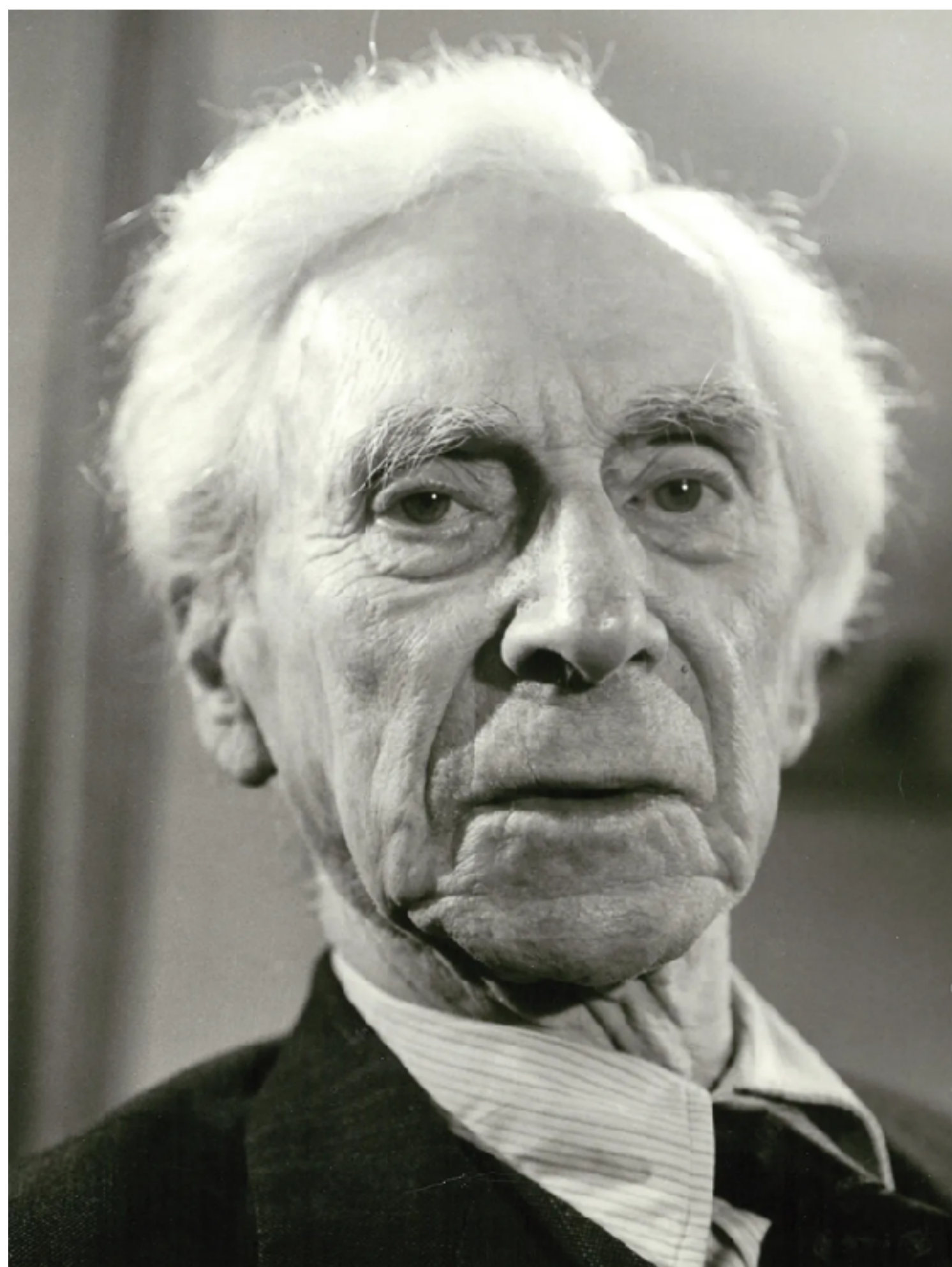
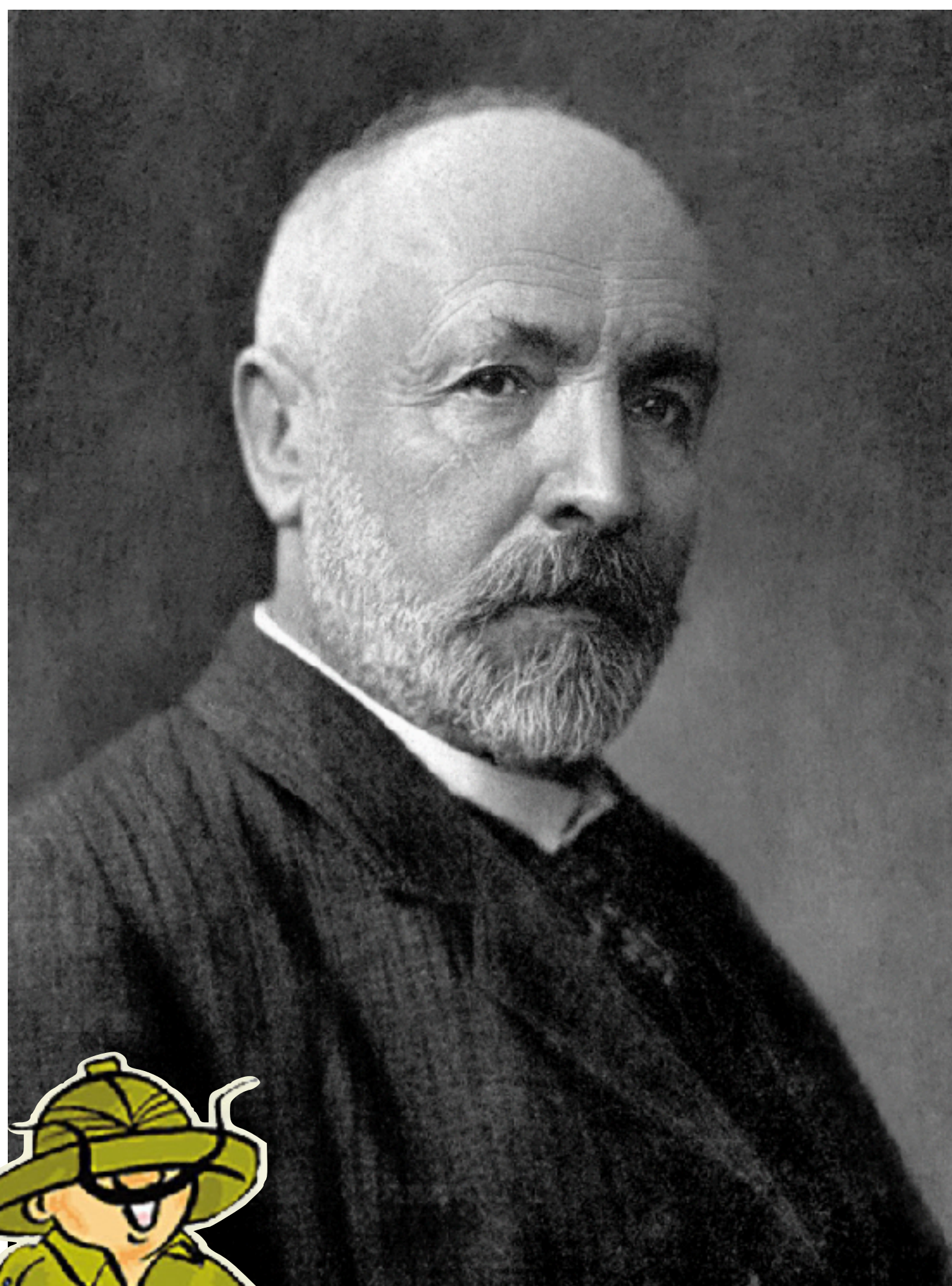
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No method of computation can outsmart the diagonal!





Don't go mad!

